

13 – End-to-End Use Case (one request from start to finish)

Story: Priya Sharma, a process engineer in Fremont, needs a plasma etch uniformity test because customer X reported 4% non-uniformity. Let us follow her request **TV-2026-00015** through all 6 phases and see what the system does at each step.

Phase 1 – Request Initiation (12-Oct-2026)

Priya opens the app → **New Request:**

Field	Value
Title	Plasma etch uniformity test on 300mm wafers
Request Type	New
Department	Process Engineering
Location	Fremont, CA
Cost Center	CC-4100 Etch R&D
Priority	High
Required By	15-Dec-2026
Estimated Budget	USD 42,000
Business Justification	Customer X reports 4% within-wafer non-uniformity on 3nm etch. Need to qualify new recipe.
Technical Description	25 wafers, 3 recipes, CD-SEM measurement at 49 points per wafer.
Expected Outcome	Uniformity below 2%
Attachment	Etch_Test_Spec_v1.pdf

She clicks **Submit**.

System does:

1. App validates fields + attachment  → saves the record → TVID **TV-2026-00015**.
2. F09 uploads the PDF to `TV Requests/TV-2026-00015/02-Technical Documents`.
3. Status → **Submitted**; history row "Draft → Submitted".
4. F01 creates the 6 folders, saves the folder URL, emails Priya (**acknowledgement**) and the APPG mailbox (**new request**).

Phase 2 – APPG Review (13–16 Oct)

- **13-Oct, David (APPG)** opens the APPG Queue → TV-2026-00015 → **Start Review**, APPG Owner = David. → Status **Under APPG Review**; Priya gets "review started" email.
- David notices the wafer map is missing → comment *"Please attach the wafer map for the 25 wafers."* → **Request More Info**. → Status **More Info Required**; Priya gets email with the comment.
- **15-Oct, Priya** uploads `Wafer_Map.xlsx`, writes *"Wafer map attached."* → **Resubmit**. → Status **Under APPG Review**; David is notified. (*If Priya had not answered: reminder on day 3, escalation to her manager on day 5, auto-cancel on day 30 – flow F05.*)
- **16-Oct, David** is satisfied → **Move to Internal Alignment**.

Phase 3 – Internal Alignment (17 Oct – 31 Oct)

1st Internal Alignment – David asks 3 institutions for quotes and enters options:

Option	Institution	Capability	Quote (USD)	Weeks	Constraint
A	Stanford Nanofab Lab	5	39,500	10	Lab free from Nov
B	Fraunhofer IPMS	4	35,000	16	Shipping to Germany
C	Internal Lab Fremont	3	22,000	20	Tool booked till Jan

Quotes uploaded to **03-Feasibility & Quotes**. Requester gets an **alignment update** email.

2nd Internal Alignment – meeting with Priya and program manager Tom: Option **A** chosen because only it meets the 15-Dec date. David clicks **Select this option** → **Confirm Option** (status Internal Alignment 2), adds Tom as stakeholder, then **Send for Funding Approval**.

Phase 4 – Funding Approval (2–3 Nov)

F04 runs:

1. Cost center CC-4100 → owner **Maria Lopez**.
2. Amount 39,500 ≤ limit 50,000 → only one approval level.
3. Maria is added to the request's access team (can now view it).
4. Funding Approval row created (Pending). Approval appears in Maria's Outlook / Teams.

3-Nov, Maria clicks **Approve**, comment *"Approved from Q4 budget."*

→ Funding Approval row = Approved; Request **Funding Approved**, Approved Amount 39,500; Priya and David get **funding approved** email.

(If Maria had rejected with "No budget this quarter" → status Funding Rejected, Priya gets the reason; request ends.)

Phase 5 – Execution (10 Nov – 28 Feb)

David clicks **Start Execution** → status **In Execution**. He adds milestones:

#	Milestone	Planned	Actual	%
1	Wafer preparation	10–24 Nov	12–26 Nov	100
2	Etch runs (3 recipes)	25 Nov–15 Dec	27 Nov–20 Dec	100
3	CD-SEM measurement	16 Dec–20 Jan	21 Dec–25 Jan	100
4	Final report	21 Jan–15 Feb	26 Jan–20 Feb	100

- **5-Dec:** issue "Lab tool down for maintenance", severity High → **exception email** to David and Priya. Mitigation: move run to 2nd tool → Resolved 9-Dec.
- Costs entered: Lab fee 18,000 (Nov) + Lab fee 12,000 (Dec) + Materials 5,300 + Shipping 2,500 = **37,800** → Utilization **96%** of 39,500.
- Expected completion updated to 28-Feb. Status updates emailed to Priya and Tom.
- 28-Feb: David clicks **Complete Execution** → status Execution Completed.

Phase 6 – Closure (5 Mar 2027)

David uploads `Final_Report.pdf` to `06-Final Report & Closure` and fills:

Field	Value
Final Outcome	Recipe B improved uniformity to 1.6% (target < 2%) 
Lessons Learned	Book external lab 6 weeks early; keep backup tool option.
Reuse Recommendation	Reuse with changes

Clicks **Close Request** → status **Closed**, Closed On 5-Mar-2027.

F07: builds summary PDF (`TV-2026-00015_Summary_20270305.pdf`) into SharePoint, and emails Priya, David, Maria and Tom: **"TV-2026-00015 completed – final report available"** with links.

What management sees in Power BI

- Cycle time for TV-2026-00015 = 12-Oct-2026 → 5-Mar-2027 = **144 days**.
- Cost: approved 39,500, actual 37,800, variance +1,700.
- Bottleneck: most time spent in *In Execution* (expected), 2 days lost in *More Info Required*.
- Stanford Nanofab: 1 request, 100% on budget.

Full audit trail (Status History for TV-2026-00015)

When	From → To	By
12-Oct 10:05	Draft → Submitted	Priya
13-Oct 09:12	Submitted → Under APPG Review	David
13-Oct 09:40	Under APPG Review → More Info Required	David
15-Oct 14:20	More Info Required → Under APPG Review	Priya
16-Oct 11:00	Under APPG Review → Internal Alignment 1	David
31-Oct 16:30	Internal Alignment 1 → Internal Alignment 2	David
02-Nov 09:00	Internal Alignment 2 → Pending Funding Approval	David
03-Nov 15:45	Pending Funding Approval → Funding Approved	Maria (via flow)
10-Nov 08:30	Funding Approved → In Execution	David
28-Feb 17:00	In Execution → Execution Completed	David
05-Mar 10:10	Execution Completed → Closed	David